SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	G MAYOUN	**Reg. No.:**	192511222
Section / Batch:	CSE	**Date:**	13/07/2026

Code of Conduct

I G MAYOUN(192511222)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
		GRAND TOTAL:	**100 Marks**

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Marks)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

SIMATS ENGINEERING

Engineer to Excel

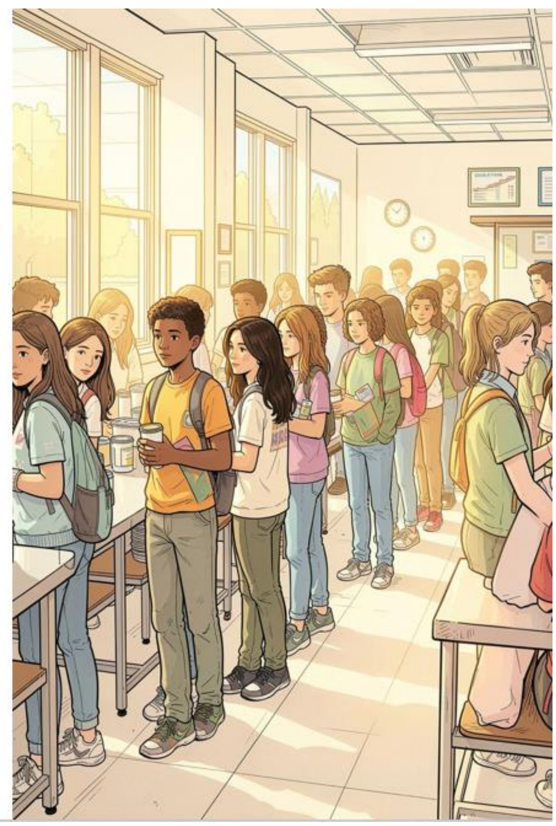
SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

A CAFETERIA TOKEN SYSTEM SERVES CUSTOMERS STRICTLY IN THE ORDER OF ARRIVAL

CSA0309-DATA STRUCTURES



GUIDED BY

Dr. Uma Priyadarsini

PRESENTED BY

G MAYOUN(192511222)

The Hidden Cost of Chaos

Lost Time

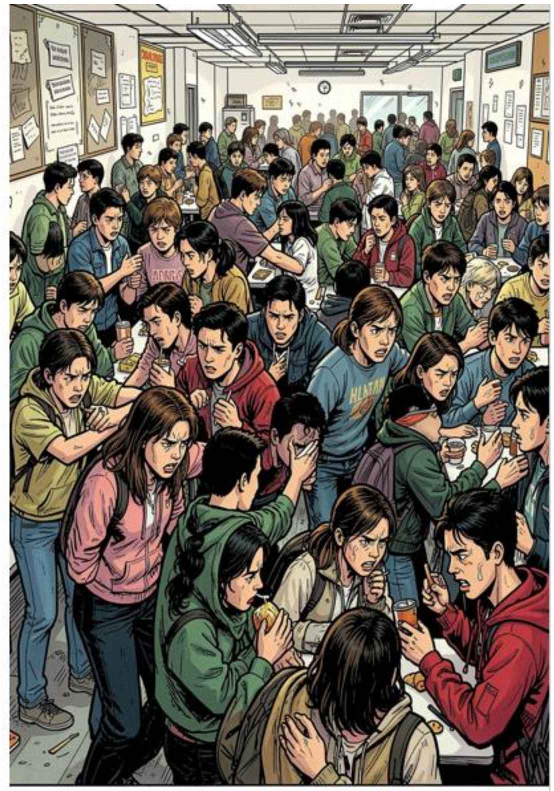
Unmanaged lines create congestion and erode break periods — students lose valuable rest or study time.

Psychological Toll

Perceived unfairness makes waits feel longer and increases frustration, reducing overall satisfaction.

Operational Risk

Chaos breeds disputes and inefficiencies — the objective is a fair system that fits existing schedules.



The Logic of Fairness: Why the Queue ADT?

Queue ADT (FIFO) — an abstract data type where elements are added at the rear and removed from the front.

- Guarantees strict order: the first arrival is the first served.
- Prevents cutting and ambiguous priority disputes.
- Simple to implement and easy for everyone to observe and trust.

How the System Works

Token Issuance

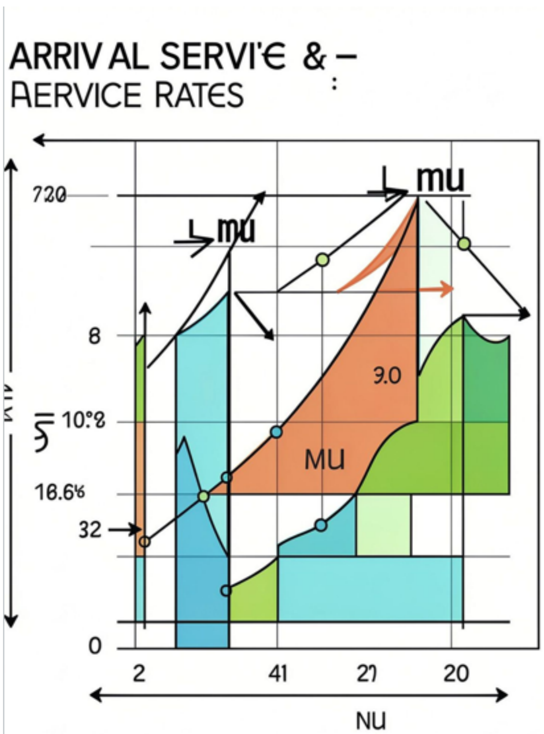
Each arrival receives a numbered token — a visible timestamp of arrival order.

Enqueue / Dequeue

Tokens are enqueued at the rear. Staff dequeue from the front, announcing numbers or displaying them on a board.

Observable Fairness

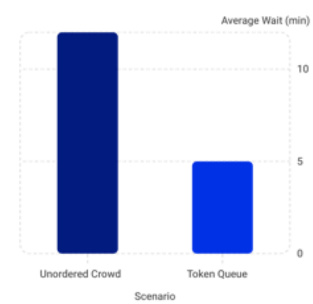
Transparent numerical order removes ambiguity and reduces disputes — no need for policing.



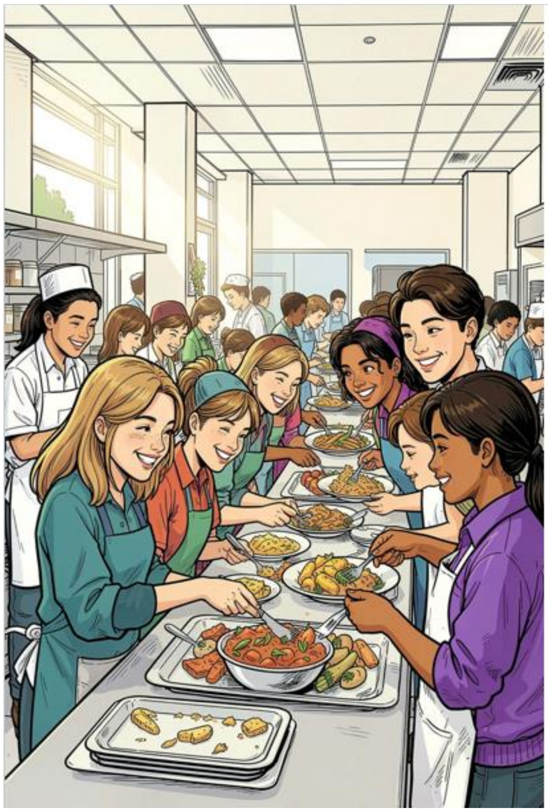
Mathematical Reliability

Model with Queuing Theory: arrivals (λ) versus service rate (μ). Use the Queue ADT to control order while capacity planning reduces bottlenecks.

- Predictable performance: compute expected queue length and waiting time for given λ and μ.
- Proactive tuning: adjust servers or service speed to keep utilisation in a healthy range.
- Empirical evidence: organised token systems reduce peak queue lengths and average wait times versus unordered crowds.



Illustrative comparison: token-based queue cuts average waits — use real data to calibrate.



Conclusion: Fairness is a Process

1

Simple

A numbered token is low-cost and intuitive.

2

Transparent

Visible order reduces disputes and perceived unfairness.

3

Reliable

The Queue ADT (FIFO) mathematically and practically enforces fairness.

Adopt the Queue ADT with monitoring of arrival/service rates to maintain efficiency and fairness.